Neuroscience Letters 576 (2014) 28–33

Contents lists available at ScienceDirect

Neuroscience Letters

journal homepage: www.elsevier.com/locate/neulet

Simple and difficult mathematics in children: A minimum spanning tree EEG network analysis

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HIGHLIGHTS

- EEG network analysis of mathematical cognition.
- Minimum spanning tree indices derived from Phase Lag Index values were used.
- We investigated the reaction of different frequencies.
- The integrated activations differed in the different frequency bands.
- Network parameters were positively correlated with math achievement.

ARTICLE INFO

Article history:
Received 4 April 2014
Received in revised form 10 May 2014
Accepted 21 May 2014
Available online 2 June 2014

Keywords:
EEG
Graphs
Minimum spanning tree
Mathematics

ABSTRACT

Sensor-level network characteristics associated with arithmetic tasks varying in complexity were estimated using tools from modern network theory. EEG signals from children with math difficulties (MD) and typically achieving controls (NI) were analyzed using minimum spanning tree (MST) indices derived from Phase Lag Index values – a graph method that corrects for comparison bias. Results demonstrated progressive modulation of certain MST parameters with increased task difficulty. These findings were consistent with more distributed network activation in the theta band, and greater network integration (i.e., tighter communication between involved regions) in the alpha band as task demands increased. There was also evidence of stronger intraregional signal inter-dependencies in the higher frequency bands during the complex math task. Although these findings did not differ between groups, several MST parameters were positively correlated with individual performance on psychometric math tasks involving similar operations, especially in the NI group. The findings support the potential utility of MST analyses to evaluate function-related electrocortical reactivity over a wide range of EEG frequencies in children.

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1. Introduction

In recent years analyses of hemodynamic and electrophysiological data using algorithms derived from graph theory have been applied extensively to the study of brain networks at rest and during performance of cognitive tasks [1–3]. While the body of research devoted to the neurobiological bases of math-related cognitive functions is rapidly rising in recent years, very few studies have examined underlying neuronal networks. In school-aged children older than approximately 9

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http://dx.doi.org/10.1016/j.neulet.2014.05.048
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years, activation patterns are similar to those observed in adults [4–6].

Tools derived from network science have been particularly useful in efforts to construct dynamic models of function-specific brain networks. In these conceptualizations network nodes are represented by voxels (or regions of interest in MRI/fMRI data), scalp electrodes or magnetic sensors, or estimated intracranial sources derived from EEG or MEG data. Despite recent progress in graph theoretical analysis, important issues remain unresolved, including methods to compare networks across experimental conditions and groups [8,9], where the number of connections per node, the number of links (edges), and the overall size of the network, may affect the results.

A relatively novel approach to graph modeling is based on minimum spanning tree (MST) indices derived from a weighted graph. MST analyses are used to define subgraphs by ensuring that all nodes are connected to each other without forming circles or loops [9–12]. The nodes of the MST are the same as in the original graph (for N nodes, there are exactly N−1 links). Graphs obtained for different populations or experimental conditions can be directly compared on the basis of quantitative indices, graphically outlined in Fig. 1A (suppl.).

Supplementary Figure related to this article can be found, in the online version, at http://dx.doi.org/10.1016/j.neulet.2014.05.048.

Very few studies have, to date, explored neurophysiological correlates of mathematical cognition using graph analysis [13–15]. Graph analysis relying on MST indices has been used sparsely to describe surface-level, electrophysiological connectivity patterns (indicative of network integration) comparing age groups [9], patients suffering from neurological disorders or healthy individuals (e.g. [7,16,17]).

The present study examined surface-level connectivity patterns, utilizing MST indices, in school-age children with typically achieving on math tests while engaged in subtraction and simple addition tasks. EEG data were also obtained from an age and verbal IQ-matched group of students demonstrating persistent difficulty in acquiring basic calculation skills through Grades 3–5. The goal of the study was to assess systematic individual variability in the outline of sensor-level neural networks as a function of task difficulty and arithmetic skill as well as to find the between the group differences. It was hypothesized that EEG frequencies known to be involved in long-range regional interdependencies (i.e., theta, alpha) would show MST parameters indicative of less integrated topology in the more difficult, subtraction task. Moreover, MST parameters in higher frequency bands (beta), presumed to be mainly involved in interregional connectivity, were expected to show MST parameters indicative of less integrated, local activity (more segregated) with increasing task demands.

2. Materials and methods

2.1. Participants

With permission from the University of Crete, School of Medicine Ethics Committee (in agreement with the Declaration of Helsinki) and the Research Department of the Greek Institute of Education, and after parents signed an informed consent, 1382 students attending Grade 3 were screened for math difficulties, using the Greek Math Screening Test (GMST, [18]) consisting of 13 simple calculation problems (Time 1). Two groups of students were selected based on their performance: a low performance group of students scoring below the 16th percentile on GMST ($n = 122$), and a typical performance group, scoring above the 35th percentile on GMST ($n = 122$). Parental consent for further cognitive assessment was obtained for 124 students who were administered a follow up cognitive and comprehensive math assessment at the beginning of Grade 4 (Time 2). Fifty one students met the inclusion criteria for the EEG study: Estimated WISC-III Full Scale IQ > 80, age-adjusted score above the 25th percentile on standardized word and pseudoword reading tests, and age-adjusted T score lower than 70 points on any of the eight Child Behavior Checklist-Parent Version syndrome scales assessing attention and behavior problems. Students in the math disability group (MD; $n = 20$, 11 boys, 9 girls) scored below the 16th percentile on the Neuropsychological test Battery for Number Processing and Calculation, whereas students in the math non-impaired group (NI; $n = 31$, 21 boys, 10 girls) had total NUCALC scores above the 50th percentile. All were right-handed with normal or corrected to normal vision. As shown in Table IA (suppl.), the two groups were comparable on gender distribution, age, and VIQ, although on average NI participants outperformed MD students on PIQ and Digit Span standard scores. EEG recordings on all students were obtained one year later (Time 3). At that time group assignment was verified for all students based on repeated administration of NUCALC.

Supplementary Table related to this article can be found, in the online version, at http://dx.doi.org/10.1016/j.neulet.2014.05.048.

2.2. Cognitive and math assessment

At Time 2 (Grade 4) the following tests were administered to students who met the screening criteria to exclude intellectual, reading and behavior problems: Vocabulary and Block Design subscales of the Wechsler Intelligence Scale (WISC-III [19]) to provide a rough estimate of PIQ and VIQ, respectively, and Digit Span; Word and Pseudoword reading subscales from the Test of Reading Proficiency (TORP [20]); Greek Math Screening Test (GMST [18]); Neuropsychological test Battery for Number Processing and Calculation (NUCALC [21]), for a more detailed assessment of math skills including mental calculation (addition, subtraction); Child Behavior Checklist-Parent Version (CBCL [22]) to assess externalizing and internalizing behavior problems. At T3 (Grades 5 and 6) all students were readministered GMST and the addition, subtraction and multiplication subscales of NUCALC (timed tests).

2.3. Electroencephalographic recordings

EEG was recorded from 29 electrode sites according to a modified international 10/20/system using an Electrocap while participants performed two math tasks (addition and subtraction) and during visual fixation (control condition). Recordings were obtained with CONTACT Precision Instruments amplifiers and digitized on a DATAQ AD Converter. The children were seated in a comfortable chair in a light- and sound-attenuated room.

The addition task involved 16 single-digit addition problems whereas the more demanding subtraction task consisted of 16 calculation problems involving subtraction of a single-digit number from a two-digit number. Each problem was presented above a two-digit number which represented either a correct or an incorrect result (one or two units greater or smaller than the correct sum) with a relative probability of 0.50. The nominal intertrial interval was set to 2 s. Participants were instructed to press one mouse key to indicate a correct equation and the other key to indicate an incorrect sum or difference. For each task 12 s of EEG were recorded with a star presented during the break between tasks when no recording was made. The EEG waveforms were monitored online by the experimenter who initiated presentation of the stimuli upon detecting an artifact-free period. Recording was paused during periods of frequent artifacts to allow participants

to re-engage in the task. The correspondence between keys and type of answer was counterbalanced across participants as was the order of conditions. All electrode impedances were kept below 5 kΩ. High- and low-pass filters were set at 0.004 Hz and 200 Hz, respectively, and sampling rate at 500 Hz. Offline, a single 8 s epoch (4096 samples) was selected per participant and condition covering a period during which the participant was engaged in three consecutive calculations and the EEG was free from visible artifacts.

Graph analyses were performed in the following frequency bands, theta (4–6.5 Hz), alpha1 (6.5–9 Hz), alpha2 (9–12 Hz), and beta (12–30 Hz), with BRAINWAVE v.9.54 (C.J. Stam, http://home.kpn.nl/stam7883/brainwave.html). Weighted graphs were constructed from Phase Lag Index (PLI) values, which calculate the asymmetry of the distribution of phase differences between two time series. The values of PLI reflect true interactions that are unaffected by field spread, volume conduction, or the presence of common sources contributing to both electrodes in a pair [23]. Graphs were constructed from PLI matrices for each frequency band using the minimum spanning tree (MST) method, which relies on a fixed number of nodes and edges, and is independent of average coupling strength [9]. The MST consists of the strongest connections in the entire weighted, undirected graph so that the sum of the weighted path length is minimized without forming cycles [16,24,25]. This method is considered as more suitable relative to the traditional small-world model in identifying crucial "hubs" in the network (i.e., nodes of particular importance for global information flow [3]).

The following parameters characterizing the MST of each participant and condition were computed: Maximum Degree (DEG: the highest number of connections formed by each node), Leaf Number (L: the number of nodes with Maximum Degree = 1). Leaf number presents an upper bound to the diameter of the tree (DIAM), which is the largest distance between any two nodes of the tree. Additional indices included Eccentricity (ECC: the longest distance between a given node and any other node, measured in number of edges), Betweenness Centrality (BC: the fraction of shortest paths between any pair of nodes that are running through a particular node), Tree Hierarchy (T_h: an estimate of the balance between diameter reduction and overload prevention), and Degree Correlation (R: an index of whether the degree of a node is influenced by the degree of its neighboring vertices). A given graph is considered as Assortative if $R > 0$, and Disassortative if $R < 0$. Betweeness Centrality, Degree Correlation and Eccentricity reflect the importance of individual nodes. Less integrated graphs are characterized by a small Leaf Number, low Maximum Degree, large Diameter, and high Eccentricity. More highly integrated graphs have a small Diameter, a large Leaf Number, and low Eccentricity.

2.4. Statistical analyses

MST indices were submitted to separate ANOVAs with group as the between-subjects variable, and repeated measures for Task (Fixation, Addition, Subtraction). MST indices that characterize individual nodes (BC, DEG, and R) in each frequency band were submitted to separate ANOVAs with group as the between-subjects variable, and repeated measures for Task (Fixation, Addition, Subtraction), Hemisphere (left, right), Area (fronto-central, temporo-parietal), and Electrode (F3/4, F7/8, FC3/4, and FT7/8 for fronto-central and T3/4, TP7/8, CP3/4, and P3/4 for temporo-parietal). The degree and direction of associations between MST indices demonstrating significant ANOVA results and performance on corresponding NUCALC tasks was evaluated through Pearson correlation coefficients.

Table 1
ANOVA results on MST indices by EEG frequency band.

Effect	MST index	
Theta (4–6.5 Hz)		
Task	Tree hierarchy	$F[2,82] = 3.720$, $p = 0.028$
		Lowest for subtraction (linear effect: $p = 0.012$)
Alpha1 (6.5–9 Hz)		
Task	Diameter	$F[2,82] = 3.771$, $p = 0.027$
		Lowest for subtraction (linear effect: $p = 0.009$)
	Eccentricity	Lowest for subtraction (linear effect: $p = 0.031$)
Alpha2 (9–12 Hz)		
Task	Diameter	$F[2,82] = 3.644$, $p = 0.030$
		Lowest for subtraction (linear effect: $p = 0.023$)
	Eccentricity	$F[2,82] = 3.757$, $p = 0.027$
		Lowest for subtraction (linear effect: $p = 0.015$)
Group	Degree correlation	$F[1,41] = 4.716$, $p = 0.036$
		MD ($-.37 \pm .02$) > NI ($-.43 \pm .018$)
Beta (12–30 Hz)		
Task	Degree correlation	$F[2,82] = 3.144$, $p = 0.048$
		Highest for subtraction (linear effect: $p = 0.014$)
	Diameter	Highest for subtraction (linear effect: $p = 0.026$)
	Eccentricity	Highest for subtraction (linear effect: $p = 0.022$)
	Leaf number	$F[2,82] = 3.570$, $p = 0.033$
		Lowest for subtraction (linear effect: $p = 0.019$)

MD: students with math difficulties, NI: typically performing students.

3. Results

3.1. Behavioral assessments

Performance on the GMST and NUCALC subscales (Table IA suppl.) showed that the NI participants significantly outperformed their MD peers on both accuracy ($p < .01$) and speed measures ($p < .0001$) and that the magnitude of these differences persisted across T2 (4th Grade) and T3 (5th or 6th Grade).

3.2. Network parameters

Table IIA (suppl.) summarizes MST parameter values for each group. Significant ($p < .05$) task effects were documented for several MST indices as shown in Table 1. These effects were characterized by linear trends across the Fixation, Addition, and Subtraction tasks. Overall, results indicated that Tree Hierarchy (theta band), Diameter and Eccentricity (alpha bands) were lowest for Subtraction and highest during Fixation. These linear trends were reversed in the beta band with highest Diameter, Eccentricity, and Degree Correlation for Subtraction, lower for Addition, and lowest for Fixation. Additionally, Leaf Numbers in the beta band were smallest during Subtraction as compared to Addition and Fixation. Additionally, there was a Group effect for Degree Correlation in alpha2 (MD > NI). Analyses assessing interactions of Task and Group with scalp location (Area) and hemisphere failed to reveal significant effects.

Supplementary Table related to this article can be found, in the online version, at http://dx.doi.org/10.1016/j.neulet.2014.05.048.

Across groups, moderate, significant correlations were found between the majority of MST indices where task effects were found, and performance on NUCALC tasks. These results are shown in Table 2. Perhaps the most notable finding was that Eccentricity and Diameter in the alpha2 band was positively correlated with Subtraction Speed (i.e., higher MST values were associated with slower performance on the most demanding math task). Moreover, Eccentricity and Diameter in the alpha1 and beta bands during the Addition task correlated positively with Addition Accuracy and negatively with time to complete NUCALC additions. Thus, higher overall Eccentricity and Diameter of the surface network were associated with better performance on the least challenging task. The

Table 2
Pearson correlation coefficients between MST indices and calculation performance on NUCALC by EEG band.

NUCALC index	Addition-accuracy	Addition-speed	Subtraction-accuracy	Subtraction-speed
MST index				
Alpha1 (6.5–9 Hz)				
ECC	0.45** (0.26/0.51*)	−38** (−0.27/−0.31)	<0.1	<0.1
DIAM	0.47** (0.24/0.54*)	−0.39** (−0.29/−0.29)	<0.1	<0.1
Alpha2 (9–12 Hz)				
ECC	<0.1	<0.1	<0.1 (−0.44*/<0.1)	0.38** (0.63*/<0.1)
DIAM	<0.1	<0.1	<0.1 (−0.45*/<0.1)	0.37* (0.57*/<0.1)
Beta (12–30 Hz)				
ECC	0.35* (0.34/<0.1)	−0.38** (−0.66**/<0.1)	<0.1	<0.1
DIAM	0.32* (0.30/<0.1)	−0.39** (−0.69**/<0.1)	<0.1	<0.1
DEG	−0.33* (<0.1/0.51*)	0.13 (<0.1/<0.1)	<0.1	<0.1
L	−0.37** (−0.38*/<0.1)	0.25 (0.63**/<0.1)	<0.1	<0.1

Correlation coefficients were computed for corresponding tasks (e.g., ECC during EEG addition task and NUCALC Addition). In parentheses, coefficients computed separately for the NI/MD groups.
* $p < .01$.
** $p < .001$.

opposite trends were found for Maximum Degree and Leaf Number in the beta band (better NUCALC addition performance was associated with lower DEG and L index values). Finally, Eccentricity and Diameter in the alpha2 band were inversely associated with performance on the NUCALC subtraction task. The MST tree estimated for each arithmetic task is displayed in Fig. 1 from a representative typically-achieving participant demonstrating the characteristic line-like organization.

4. Discussion

The present study examined aspects of brain network organization using the MST method in children with and without math difficulties during performance of arithmetic tasks that varied in processing demands. Results indicated that whenever a task main effect was found, the linear trend was also significant suggesting a progressive modulation of the corresponding parameter with increased task difficulty. Although the study of interdependency measures as indices of functional neuronal connectivity is still in its infancy, certain assumptions can be made with respect to the potential significance of these measures at the level of psychological functions. A very basic outline of the results which were evident across achievement groups can be summarized as follows:

First, Tree Hierarchy in the *theta band* was lowest during the more demanding subtraction task, indicative of a more "line-like" or less integrated topology. This is in accordance with the purported

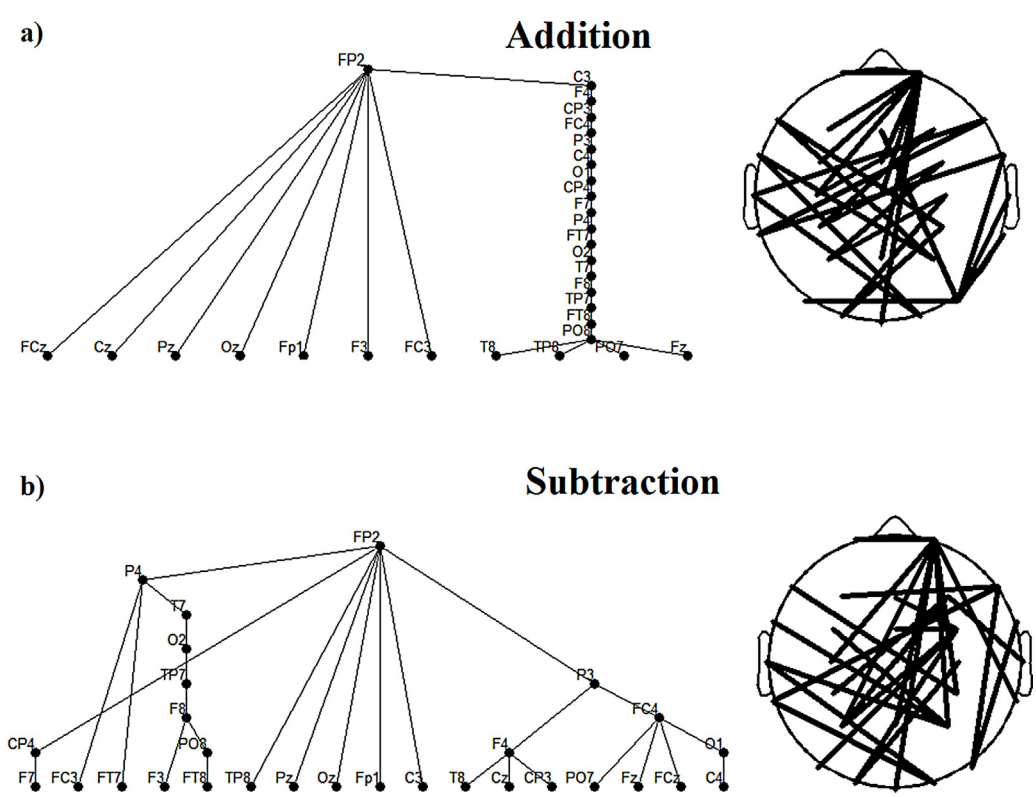

Fig. 1. Representative MST tree in the alpha2 band for an NI participant during addition and subtraction.

role of theta band oscillations in long-range cortical signaling [26], assuming that the more complex math task is likely to engage a more anatomically complex and distributed network [4].

Second, the subtraction task was associated with *lower* Diameter and Eccentricity values, compared to the addition and fixation conditions, in the *alpha1 and alpha2 bands*. Taken together these findings reveal a less "line like" (or more "star-like") topology during subtraction in these bands. Increased activity in alpha1 and alpha2 bands is often found during cognitive tasks, usually in combination with activity in higher frequencies [27]. Long-range synchronizations between neuronal assembles have been described in several studies during engagement of cognitive processes, known to be involved in demanding calculation tasks, such as working memory [28,29]. In contrast to the theta band, MST parameters in the alpha bands were indicative of greater network integration as task demands increased. In particular, Diameter and Eccentricity estimates may be suggestive of a network that contained a greater number of nodes crossed by shortest paths, thus more centralized and integrated in overall layout. Importantly, correlational results indicated that the more centralized/integrated the network in the alpha bands was the more efficient the performance on the timed NUCALC subtraction task.

Third, the pattern of task-related changes in MST parameters was very distinct in the higher frequency range *(beta band)*, which is generally considered as mainly involved in intraregional connectivity [30–32]. In this band, performance of the more demanding subtraction task was associated with *higher* diameter, eccentricity, and Degree Correlation values as compared to addition and fixation. This pattern of MST values suggests more line-like local networks, further demonstrating a high degree of assortativity (i.e., greater tendency for nodes showing similar connection patterns to be interconnected).

A puzzling outcome of the present study was the failure to find significant differences between age- and IQ-matched achievement groups, which may be accounted for in several ways. For instance, it is possible that regional dysfunction reported previously in MD children during performance of math tasks by fMRI and MEG studies is not accompanied by functional altered connectivity detectable by surface EEG recordings. It is also possible that the task-related differences found in the present study more closely reflect task variations in overall difficulty (and the associated modulation of the degree of engagement of attentional and working memory processes), rather than differential engagement of processes specialized for arithmetic calculations.

Results were in general agreement with expectations regarding the roles of oscillations in different frequencies in regional brain connectivity. There we a number of methodological limitations. For instance, connectivity patterns were assessed at the sensor level, preventing a detailed assessment of cortico-cortical interdependencies and corresponding network outlines. Moreover, PLI was computed on relatively long EEG segments which were not precisely time-locked to the presentation of calculation problems. This method was adopted in order to include a wide set of cognitive operations involved in calculation in view of the highly variable time frame for completing each problem by school-aged children, especially those experiencing math difficulties. Although this choice may have affected the generalizability of the results it will be necessary to compare our results with MST indices derived from stimulus-time-locked data.

5. Conclusions

The present study represents one of the first attempts to establish links between cognitive task demands and brain network features in the form of MST parameters. A potentially promising research direction would target network hubs which, in line with our findings regarding assortativity, may tend to be more closely connected to other centrally located hubs, in response to heightened processing demands. The existence of such hubs has been recently postulated on the basis of Diffusion Tensor Weighted data [33] as a characteristic feature of a "rich-club" brain organization. Future studies exploring this network feature would probably need to take into account connectivity patterns in multiple frequency bands, concurrently, estimated in real time, i.e., during the unfolding of the targeted cognitive function, in order to isolate network features that are more intimately linked to task component processes.

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